Abhigyan Majumdar

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About Me

I am a Ph.D. student in Mechanical and Aerospace Engineering at the University of California Davis with a specialization in System Dynamics and Controls. My research focuses on developing new frameworks for capturing uncertainty in globally linear representations of nonlinear systems. I have authored several peer-reviewed publications and actively participate in interdisciplinary research and academic teaching. I am keen to contribute to the academic community through journal reviewing and collaborative research.

Education

Ph.D. **University of California Davis**, Mechanical & Aerospace Engineering — Sept 2021 – Present

- GPA: 3.92/4.0
- **Relevant Coursework:** Data-driven modeling and controls, Optimization and Learning based Control, Guidance Navigation & Control of Unmanned Aerial Systems, Uncertainty Analysis, Estimation Theory & Applications,

B.Tech **National Institute of Technology Rourkela**, Mechanical Engineering — Jul 2017 – May 2021

- GPA: 8.97/10

Skills

Languages: MATLAB, Python (NumPy, SciPy, PyTorch, Pandas, GPy, Matplotlib, CANlib), C++

Software & Tools: Simulink & Simscape, Simulink RealTime, ANSYS Fluent, TwinBuilder, Vector CANalyzer, Arbin MITS Pro

Techniques: Neural Networks, Model Predictive Control, Hardware-in-Loop Testing, Reduced Order Modeling, System Identification, Koopman Operators

Publications

Sliding Mode Wheel Slip Control for Regenerative Braking of an All-Wheel-Drive Electric Vehicle — Jan 2024

Abhigyan Majumdar, Shima Nazari

DOI:10.1115/1.4064346

This paper presents a two-step hierarchical brake controller for a dual-motor electric vehicle, based on a novel formulation of the sliding mode controller for independent axle slip regulation with a regenerative-split controller to maximize energy recovery. By addressing nonlinear tire dynamics, weight transfer, and disturbances, the proposed method outperforms traditional SMC formulations in model-in-loop simulations.

Control and Control-Oriented Modeling of PEM Water Electrolyzers: A Review — Sept 2023

Abhigyan Majumdar, Meridian Haas, Isabella Elliot, Shima Nazari

DOI:10.1016/j.ijhydene.2023.04.204

This review surveys control-oriented modeling frameworks for PEM water electrolyzers, detailing electrochemical, thermal, mass transport, and equivalent circuit models. It further examines a broad range of control techniques to provide insights on relative strengths and future research directions.

Thermal Reduced Order Modeling for System Analysis of EV Battery — Apr 2023

Peiran Ding, Weiran Jiang, ***Abhigyan Majumdar***, Pranav Pawar, Xiao Hu, Anil Wakale

DOI:10.4271/2023-01-0931
This study develops a 3-part coupled electro-thermal system model of EV battery modules to reduce computation time compared to 3D CFD simulations. An equivalent-circuit cell model is bidirectionally coupled with an LTI thermal reduced order model and a LPV cooling plate model to accurately capture critical thermal interactions, enabling efficient evaluation of various control strategies.

Analytical Study of Phase Change Heat Transfer in Biological Tissue using Coordinate Transformation Technique during Cryosurgery May 2021

Abhigyan Majumdar, Sumit Kumar

DOI:10.1615/TFEC2021.bio.036282
This study presents a novel coordinate transformation technique that converts the complex nonlinear PDEs of phase change heat transfer during cryosurgery into simpler ODEs. The resulting analytical solution, efficiently tracks the freezing front propagation and thermal history in biological tissues, providing a fast and accurate tool for optimizing cryosurgery parameters.

Awards

First Place Propulsion Controls & Modeling Presentation Award, EcoCAR EV Challenge Year 2 May 2024

- Presented to industry professionals from General Motors, MathWorks, US Department of Energy, Bosch, dSPACE and more regarding development and validation of EV propulsion controls software. Ranked first place amongst 15 universities across North America.

Over the HIL Award, EcoCAR EV Challenge Year 2 May 2024

- Ranked first place among 15 universities across North America for the best Hardware-in-the-Loop setup built using Speedgoat hardware for testing and validation of EV Propulsion Supervisory Controller code.

Experience

Graduate Student Researcher, UC Davis CORE Lab Sept 2024 – Present

- Developing data-driven algorithms for probabilistic Koopman Operator identification in nonlinear dynamic systems.
- Designing predictive safety filtering for thermal derating of EV drive units.
- Collaborating across interdisciplinary teams to validate modeling and control strategies.

Graduate Teaching Assistant, University of California Davis Jan 2022 – Present

- Conduct discussion sessions, hold office hours, conduct laboratory demonstrations, grade assignments, and assist in creation and grading of exams.
- **Courses:** *Thermodynamics, Dynamics, Vehicle and Transportation Technology, Manufacturing Processes for Biomedical Engineering, General Chemistry*

Propulsion Controls & Modeling Lead, UC Davis EcoCAR Team Sept 2022 – Aug 2024

- Led a multidisciplinary team of 15 undergraduates to revamp the powertrain of a Cadillac LYRIQ for the EcoCAR EV Challenge
- Led propulsion supervisory controller development from scratch using model-based systems engineering workflows in MATLAB-Simulink powered MIL, HIL, and VIL environments
- Implemented functional safety by focusing on hazard analysis, requirements traceability, code coverage and multi-level testing of all propulsion controls features

Battery Modeling Intern, Farasis Energy USA — Jun 2022 – Sept 2022

- Worked on reduced order modeling and coupled electro-thermal simulation of Lithium-ion battery modules using ANSYS Fluent and Twinbuilder
- Established workflow for system-level simulations that can evaluate different fast-charging and thermal control algorithms based on battery performance

Undergraduate Researcher, Bio-Heat Transfer Laboratory, NIT Rourkela — Aug 2020 – May 2021

- Developed a combined analytical-numerical method for phase change heat transfer in biological tissue during cryosurgery to preserve accuracy while reducing simulation time.

Projects

Energy Optimal Dual Stage Adaptive Cruise Controller for EVs — Apr 2023 – Jun 2023

- Developed a nonlinear model predictive controller to optimize energy efficiency and passenger comfort.
- Demonstrated real-time feasibility using MATLAB's fmincon solver.

Autonomous Aerial Landing of Quadcopter — Apr 2022 – Jun 2022

- Built a MATLAB-Simulink simulation environment for UAV autonomous landing on a moving target.
- Developed a trajectory generation algorithm and implemented adaptive LQR control.

Robust MIMO Controller for EV Cooling System — Apr 2022 – Jun 2022

- Designed a MIMO control system using Youla Parameterization and H-Infinity techniques.
- Achieved robust disturbance rejection for integrated EV cabin, battery, and motor cooling.